combined equations of general relativity and quantum mechanics have gone haywire.

Scientists have been aware of the tension between general relativity and quantum mechanics for more than half a century, but for a long time relatively few felt compelled to search for a resolution. Instead, most researchers used general relativity solely for analyzing large and massive objects, while reserving quantum mechanics solely for analyzing small and light objects, carefully keeping each theory a safe distance from the other so their mutual hostility would be held in check. Over the years, this approach to détente has allowed for stunning advances in our understanding of each domain, but it does not yield a lasting peace.

A very few realms—extreme physical situations that are both massive and tiny—fall squarely in the demilitarized zone, requiring that general relativity and quantum mechanics simultaneously be brought to bear. The center of a black hole, in which an entire star has been crushed by its own weight to a minuscule point, and the big bang, in which the entire observable universe is imagined to have been compressed to a nugget far smaller than a single atom, provide the two most familiar examples. Without a successful union between general relativity and quantum mechanics, the end of collapsing stars and the origin of the universe would remain forever mysterious. Many scientists were willing to set aside these realms, or at least defer thinking about them until other, more tractable problems had been overcome.

But a few researchers couldn’t wait. A conflict in the known laws of physics means a failure to grasp a deep truth and that was enough to keep these scientists from resting easy. Those who plunged in, though, found the waters deep and the currents rough. For long stretches of time, research made little progress; things looked bleak. Even so, the tenacity of those who had the determination to stay the course and keep alive the dream of uniting general relativity and quantum mechanics is being rewarded. Scientists are now charging down paths blazed by those explorers and are closing in on a harmonious merger of the laws of the large and small. The approach that many agree is a leading contender is *superstring theory* (Chapter 12).